

Unit 10, Lesson 7: Analyzing Descriptions and Graphs

Benchmark

MA.B.F.1.3 Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing or constant.

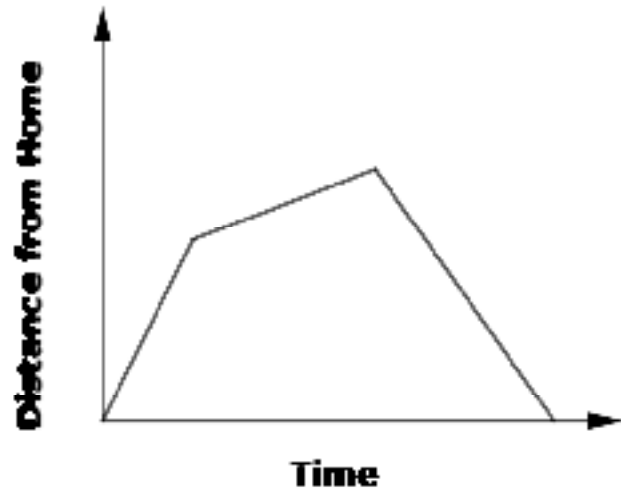
Learning Targets

- ☐ I can analyze a graph representing a real-world functional relationship.
- ☐ I can analyze a written description of a real-world functional relationship.

10.7.1 Warm-Up: Bell Work

Three stories are given.

One of the stories matches the graph shown.



Story A	Story B	Story C
Yadiel rode his bike toward his friend’s house. After riding for a few minutes, he started riding faster until he remembered he forgot his phone, so he turned around and rode home to get it.	Yadiel and his brother rode their bikes up the hill on their street. The hill is steep before it flattens out. Once they got to the top, they rode smoothly down the other side to their neighbor’s house.	Yadiel rode his bike quickly to meet up with his neighbor. Once they met up, they rode a little bit slower to meet up with another friend before turning around and riding back to Yadiel’s house to hang out.

- Determine which story matches the graph.
- Describe which clues from the graph or the story you used to match the story and the graph.

10.7.2 Activity: Card Sort

Work with your partner on the following.

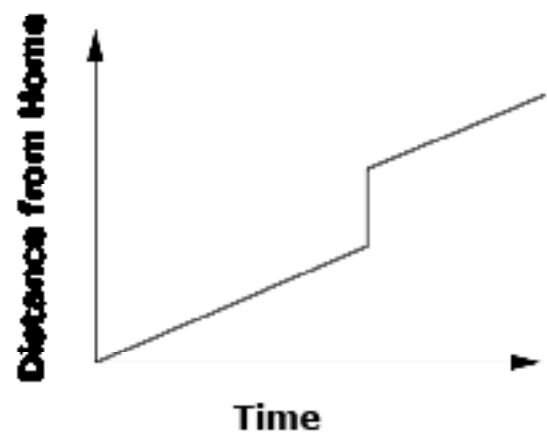
1. Match the cards. Write the number of the interpretation that matches the graph.

Graph	Interpretation
A	
B	
C	
D	
E	
F	

2. For the graph that matches Interpretation Card 6, write a story that matches this graph.

3. A graph that shows the distance from home versus time is shown.

Alexandra says the relationship represented by the graph is not possible. Explain why.



10.7.3 Bring It Together: Analyzing Graphs Representing Real-World Contexts

Consider some of the graphs and stories you have analyzed over the last few lessons.

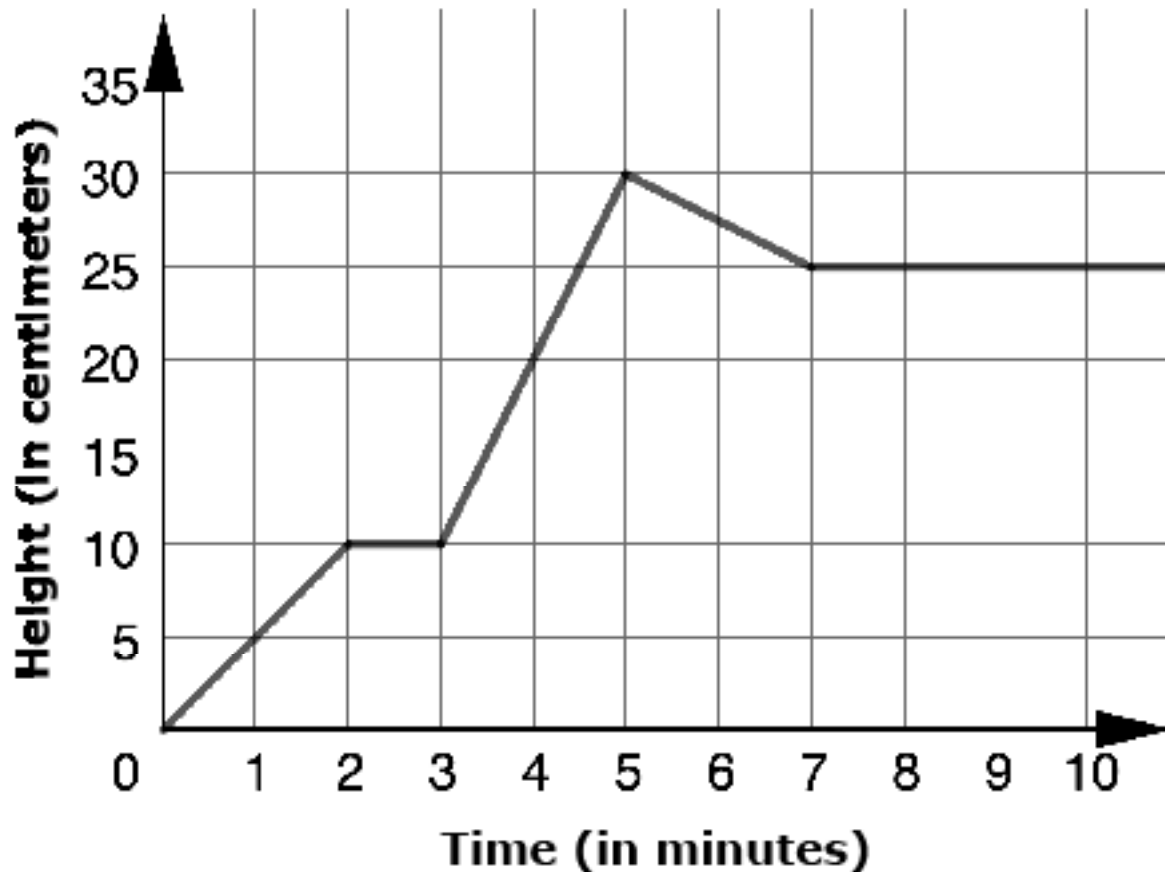
- 1. What are some strategies you have learned and key tips that are helpful for you to keep in mind when analyzing or sketching graphs of functional relationships?**

Helpful Tips	Strategies

- 2. Compare and discuss your list with your partner. Add any tips or strategies your partner shares that you find helpful to your list.**

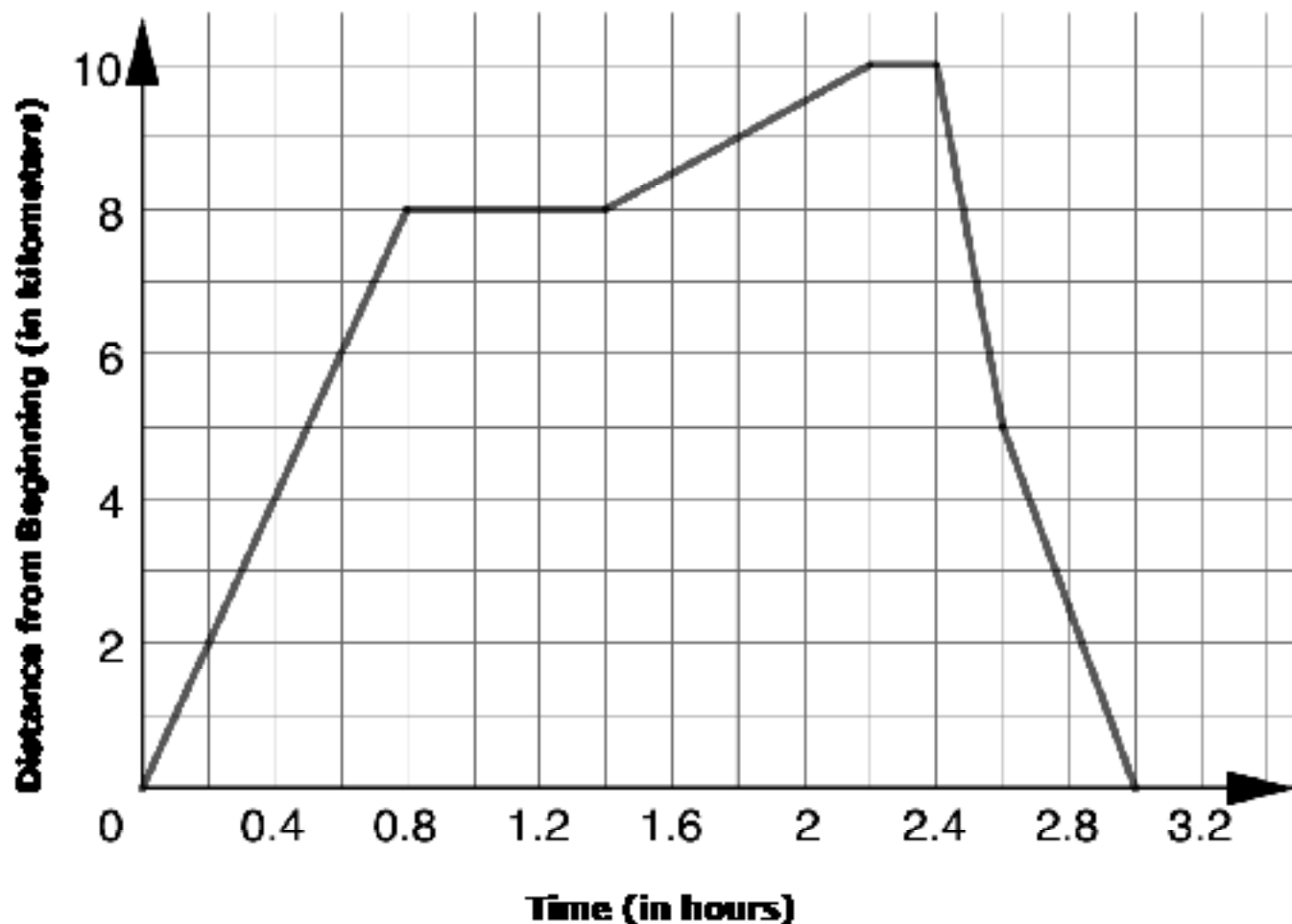
10.7.4 Your Turn!

1. Emma fills her aquarium with water. The graph shows the height of the water, in cm, in the aquarium as a function of time, in minutes.



Write a story to match the graph that describes how Emma fills the aquarium.

2. Mason was riding his bike along the Lake Okeechobee Scenic Trail. The graph shows his trip. The trail has markers every 0.5 kilometers (km), showing the distance from the beginning of the trail.



- a. What is the domain of Mason's bike ride?
- b. What is the range of Mason's bike ride?

- c. Between which times was Mason going the fastest?**

- d. Between which times was Mason riding away from the beginning of the trail?**

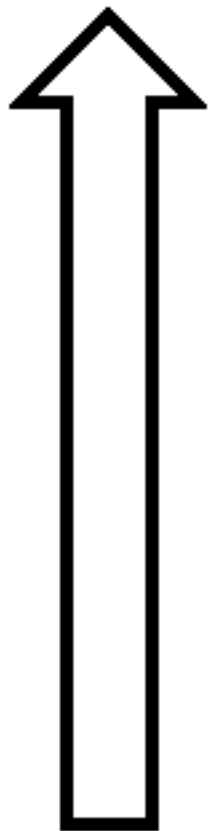
- e. Between which times was Mason riding back to the beginning of the trail?**

- f. Between which times was Mason stopped?**

10.7.5 Wrap-Up: Reflection

1. **Color in the arrow up to the statement that best describes your current understanding of the benchmark below.**

Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing or constant.



I'm so confident, I could explain how to analyze or sketch graphs of functional relationships to someone else!

I can get to the right answer, but I don't understand well enough to explain how to analyze or sketch graphs yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.